Shivin Rajesh Anand

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Education

University of Florida **Gainesville, FL**
B.S. Computer Engineering *May 2030*

Skills

FPGA & Architecture: Verilog, SystemVerilog, RTL Design, Computer Architecture, Embedded Systems, Vivado, Vitis
Programming: Python, C++, Swift, Java, JavaScript, Embedded C, Git, Linux/UNIX, Data Analysis, OpenCV, three.js, WebXR
Hardware & Circuits: PCB Design (KiCad), Board Layout, Circuit Simulation (Multisim), Debugging & Validation, CAN, I2C
CAD & Prototyping: Autodesk Inventor, Fusion 360, AutoCAD, 3D Printing, Rapid Prototyping, Mechanical Design

Experience

University of Florida, Dept. of ECE **Gainesville, FL**
Undergraduate Research Assistant, Smart Systems Lab *August 2026 – Present*

- Designing a CPU in Verilog and SystemVerilog for an FPGA/SoC project under PhD student Peter Forcha, implementing 3 ALU operations (add, subtract, multiply) and the datapath in Vivado and Vitis; learning Vivado synthesis and timing analysis
- Laying the RTL base for a quantized neural-network inference accelerator on an FPGA, following the lab's multi-tenant research
- Targeting low-latency, near-sensor ML inference as the next phase, building on the lab's near-sensor image processing research

IEEE Student Branch, University of Florida **Gainesville, FL**
Hardware Team Member – SoutheastCon Robotics Competition *September 2026 – Present*

- Designing and laying out the single custom KiCad PCB that will control all electrical systems on the team's autonomous robot for the 2027 Stock Car Race (3 laps on a 4x8 ft oval track in 3 minutes), with breadboard prototyping before fabrication
- Serving as sole electrical/PCB designer on a 10-person team, owning hardware design and validation within the 12 in cube start envelope while coordinating hardware-software system integration with the software subteam

University of South Florida, Dept. of EE **Tampa, FL**
Head Researcher, Bio-Organic Electronics Lab *April 2024 – May 2026*

- Enabled real-time monitoring of 4 environmental variables (CO2, VOCs, temperature, humidity) by designing and multimeter-debugging a KiCad PCB with a 3.3V I2C bus (BME680 + Teensy 4), then validating a 72 h log (±3 °C, ±6% RH)
- Eliminated destructive harvests by building a closed-loop hydroponic chamber (nutrient recirculation, relay-switched LED, top-down camera) in C++, run over 6 growth trials on a platform picked by decision matrix from Teensy, ESP32 and Arduino

Projects

Re:collection (Alzheimer's Memory Palace) | *Swift, ARKit, Three.js, Claude API, Team of 3* **github.com/shividoge/re-collection**

- Won 1st place in the Meta track (Bringing People Closer Together with AI) at Cal Hacks 13.0 with a 3D Memory Palace that turns family visits and old videos into rooms a person with Alzheimer's can walk through with a voice guide
- Built the iOS capture app in Swift and ARKit to record LiDAR depth and pose with video and audio, exporting 1 cm point clouds
- Shipped the pipeline from iPhone capture to a browser viewer with 200+ automated tests across the app, backend and website

Deja View (Photos to 3D Worlds) | *three.js, WebXR, COLMAP, Depth Anything V2, Team of 4* **deja-view.photo**

- Won 1st place in both the AI/ML and DeepSpace tracks at SwampHacks XII with a pipeline that turns photos or video of a real place into a walkable 3D world, viewable in a browser, WebXR headset or Roblox
- Rebuilt a 25-photo, 2-clip scene from 305 camera views into 844,875 splats, and a kitchen from 4 listing photos at measured scale (counters 0.87 m, fridge 1.85 m)

S^3 Smart Glasses (AR) | *ESP32, OLED Display, Gemini API, 3D Printing, Team of 3 (High School)* **shivin.me/smart-glasses**

- Contributed to circuit design on AR glasses whose ESP32 returns Gemini answers on a 64x32 OLED display in a mean 1,315 ms (3 s limit), while the build stayed at about $83 in parts and 86.66 g
- Characterized power and thermals over 5 trials: 6.1 h of battery life on a 200 mAh Li-ion cell and a 39.5 °C peak temperature

Autonomous Plant Biomass Estimation via Computer Vision | *Python, OpenCV, Independent Research (USF)* **Project Site**

- Achieved 6.9% MAPE and R squared of 0.91 across 6 growth trials with a Python and OpenCV pipeline (HSV conversion, Excess Green Index segmentation, morphological refinement) that extracts canopy area, then converted it to fresh biomass with a linear regression calibrated on harvest ground truth
- Earned a NASA Kennedy Space Center Award at the Florida State Science and Engineering Fair

Leadership

Middleton Robotics **Tampa, FL**
President *May 2024 – May 2026*

- Returned the school's FRC program to competition after years of inactivity, earning 2 national awards and 1 World Championship award, by rebuilding the mechanical and electrical pipeline (C++ on VEX controllers, Java and CAN bus on RoboRIO, Git)
- Expanded competitive participation 30% by coordinating 9 teams across 15+ regional competitions

Florida Student Association of Mathematics **Tampa, FL**
Region 4a Coordinator & Webmaster *June 2023 – May 2026*

- Cut manual communication overhead 60% by building the organization's website (HTML, CSS, JavaScript, Git, Linux hosting)
- Grew math-competition engagement 50% by creating paths for 80 Tampa Bay students to HMMT, PUMaC, ARML, and CMIMC

Certifications

Professional: Linux Essentials (LE-1) | Autodesk Certified User (Inventor, AutoCAD, Fusion 360) | PMI Project Management Ready
Robotics & Engineering: RECF Robotics, Programming, Electrical, Mechanical, Fundamentals of Engineering